(Please write your Exam Roll No.) Exam Roll No. .....

# END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

**Paper Code: ES-107** **Subject: Electrical Science**

**Time: 3 Hours** **Maximum Marks: 60**

***Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.***

Q1 Attempt all questions:-

(a) Distinguish between dependent and independent sources. How do you transform a voltage source into current source? **(2)**

(b) Show that power consumed in a purely inductive circuit is zero when sinusoidal voltage is applied across it. **(2)**

(c) Why a series resonant circuit is called as an acceptor circuit and a parallel resonant circuit as a rejector circuit? **(2)**

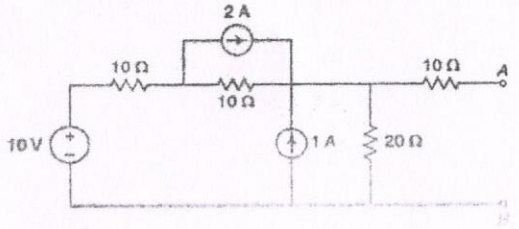
(d) Why is induced emf in a dc motor called the back or counter emf? **(2)**

(e) What is the difference between an ideal and practical transformer? **(2)**

(f) What type of damping is provided in PMMC instruments? **(2)**

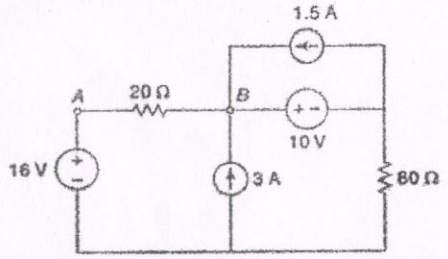
## UNIT-I

Q2 (a) Find the Thevenin's equivalent circuit at terminals AB for the given network and also determine the power dissipated in a 5 Ω resistor connected between A and B. **(7)**




(b) Derive the expression for converting a star network to a delta equivalent network. (5)

Q3 (a) Using the superposition theorem, find the voltage across the 20 Ω resistor of the given circuit. **(7)**




(b) State and explain Norton's theorem. Show that Thevenin's and Norton's theorems are dual to each other. **(5)**

**P.T.O.**



## UNIT-II

Q4 (a) A series circuit consists of a capacitor and a coil takes a maximum current of 0.314 A at 200 V, 50 Hz. If the voltage across the capacitor is 300 V at resonance. Determine the capacitance, inductance, resistance and Q of the coil. **(7)**

(b) Define parallel resonace with suitable phasor diagram. Calculate the resultant current and quality factor at resonance in terms of the pameters of a circuit. **(5)**

Q5 (a) A choke coil having a resistance of 10 Ω and inductance of 0.05 H is connected in series with a condenser of 100 μF. The whole circuit has been connected to 200 volt, 50 Hz supply. Calculate: **(7)**

(i) Impedance (ii) Current (iii) Power factor (iv) Power input (v) Apparent and Reactive power of circuit.

(b) Define quality factor in a series R-L-C circuit. Determine the half power frequencies in terms of quality factor and the resonant frequency for series R-L-C circuit. **(5)**

## UNIT-III

Q6 (a) A 220 V, dc shunt motor takes a line current of 20 A. Resistance of shunt field winding is 200 Ω and resistance of the armature is 0.3 Ω. Find the armature current and back emf. **(7)**

(b) Derive the emf equation of a dc generator. Draw and expain different types of dc generator with suitable diagram. **(5)**

Q7 (a) A 3-phase, 50 Hz induction motor has 6 poles and operates with a slip of 5% at a certain load. Determine. **(7)**

(i) The speed of the rotor with respect to the stator.

(ii) The frequency of rotor current

(iii) The speed of the rotor magnetic field with respect to rotor.

(iv) The speed of the rotor magnetic field with respect to stator.

(b) What are the various methods of starting the single phase induction motor? Explain them in brief with relative advantages and disadvantages. (5)

## UNIT-IV

Q8 (a) In a 25 KVA, 2000/200 V transformer the iron and full load copper losses are 350 W and 400 W respectively. Find the efficiency at unity power factor at (a) full load (b) half load. Determine the load for maximum efficiency. **(7)**

(b) Explain the working principle of Attraction type and Repulsion type moving iron instruments with the help of suitable diagrams. **(5)**

Q9 (a) A single phase transformer has a core whose cross -sectional area is 150 $cm^2$ ; operates at a maximum flux density of 1.1 $Wb/m^2$ from a 50 Hz supply. If the secondary winding has 66 turns, determine the output in KVA when connected to a load of 4 Ω impedance. Neglect any voltage drop in the transformer. **(7)**

(b) Draw the constructional features of transformer and explain the working in detail. Also derive the emf equation of transformer. **(5)**

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